

E.1 / PAPER 1 / QUESTION 17

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17. In the Geiger–Marsden–Rutherford experiment, alpha particles were accelerated through a potential difference of 7.7 MV and incident on a thin foil of gold. The proton number of gold is 79.

What is the distance of closest approach?

A. $\frac{158ke}{7.7 \times 10^6}$

B. $\frac{158ke^2}{7.7 \times 10^6}$

C. $\frac{79ke}{7.7 \times 10^6}$

D. $\frac{79ke^2}{7.7 \times 10^6}$